Membership strictly lowers rank: if B is in A then the rank of B is an element of the rank of A

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-spine/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the single displayed chain with its citation to valphalowerrank is verbatim in the span.

Editorial note. The label rankmemberslower names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Hierarchy as the sets of smaller rank. Same unit, \ollabel{valphalowerrank} (source line 589). The single line of the proof instantiates the equality A subset V_rank(A) = {x : rank(x) in rank(A)} at A itself.
- Definition of rank. Same unit, the rank definition preceding the propositions: rank(A) is the least alpha with A subset V_alpha. Supplies A subset V_rank(A) for the displayed chain.

Source reference. Original reference valphalowerrank: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 589–589. Prerequisite pointer invoked at proof line(s) 608; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *If* $B \in A$*, then* $\operatorname{rank} B \in \operatorname{rank} A$.

Proof. $A \subseteq V_{\operatorname{rank} A} = \{\, x \mid \operatorname{rank} x \in \operatorname{rank} A \,\}$ by valphalowerrank. □